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F78AB
Actuarial and Financial Mathematics

Chapter 9: Measuring Fund Performance

LEADERS IN IDEAS AND SOLUTIONS

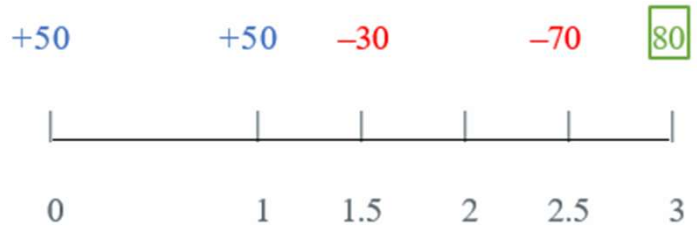
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Measuring Fund performance

- Investors are interested in how the rate of return earned by their fund manager compares with that earned by other managers or on other investments.
- We can use the **yield** as one measure. The yield is usually known as the money-weighted rate of return (MWRR) when used for this purpose.

Example

Suppose that at time $t = 0$ a fund holds investments valued at 50 and the cashflows in and out of the fund during the next three time units are as in the timeline below



and that at time $t = 3$ the fund holds investments valued at 80

We can set up an equation of value and use a calculator or Goal seek to find that the yield is $i = 0.0856$.

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Example

Consider a fund with two members, A and B.
At $t = 0$ and $t = 3$, it has nothing in it.

Member A puts in £100 at time $t = 0$ and takes out £200 at time $t = 3$.

Member B puts in £1 million at time $t = 1$ and takes out £1 million at time $t = 2$.

Consider the yield for both investors separately, and then collectively. What can we learn from these results?

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Linked Internal Rates of return

If we calculate the yield over sub-periods separately, and then link them, we can find the yield over the whole period.
This is called a **linked internal rate of return (LIRR)**

Return to *example 4.5.1*, with the following new information:

- fund value at time 1, before the cashflow of +50, is 105;
- fund value at time 2 is 135.

Calculate the yield for each sub-period, and then the linked internal rate of return.

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Comments on the LIRR

- Not uniquely defined – depends on the sub-division of time periods used for the calculation
- Provides more information than the MWRR
- With properly chosen sub-periods is much less influenced by large cashflows than MWRR

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Time-Weighted Rate of Return (TWRR)

- If we know the fund size immediately before every cashflow to and from the fund, we can simplify the process because there are no intervening cashflows to worry about.
- We calculate a **growth factor** for each time period.
- The **time-weighted rate of return (TWRR)** is the geometric average rate of return per time unit, based on the total growth factor.
- We can calculate the TWRR for the previous example

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Attendance matters!

